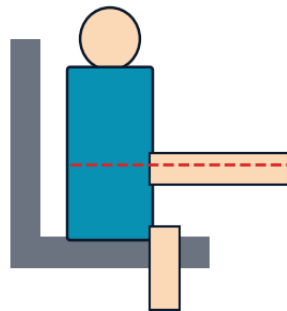


Blood Pressure Measurement Guide

Evidence-based protocol for accurate non-invasive blood pressure measurement

Correct Patient Positioning



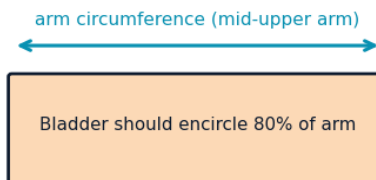
- Rest seated 5 min before measuring
- Back supported, legs uncrossed
- Feet flat on the floor
- No talking, caffeine or smoking 30 min prior

Step 1: Prepare the Patient

Have the patient rest seated for at least 5 minutes before measurement. Legs should be uncrossed, feet flat on the floor, back supported. Ask the patient to avoid talking, caffeine, smoking, or exercise for 30 minutes prior. Ensure the patient has an empty bladder.

Note: Anxiety and pain can temporarily raise blood pressure by 10–40 mmHg.

Selecting the Correct Cuff Size



Cuff sizing	
Small adult	22–26 cm
Adult	27–34 cm
Large adult	35–44 cm
Thigh cuff	45–52 cm

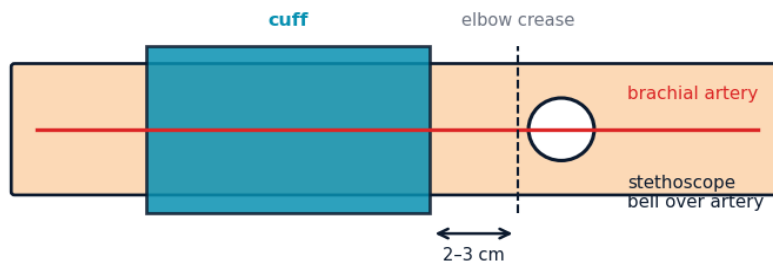
Wrong cuff size is the most common source of measurement error.

Step 2: Select the Correct Cuff Size

Measure arm circumference at mid-point between shoulder and elbow. Cuff bladder should encircle 80% of arm circumference. Use: small adult (22–26cm), adult (27–34cm), large adult (35–44cm), thigh cuff (45–52cm). An undersized cuff gives falsely high readings; oversized gives falsely low readings.

■ Using wrong cuff size is one of the most common sources of blood pressure measurement error.

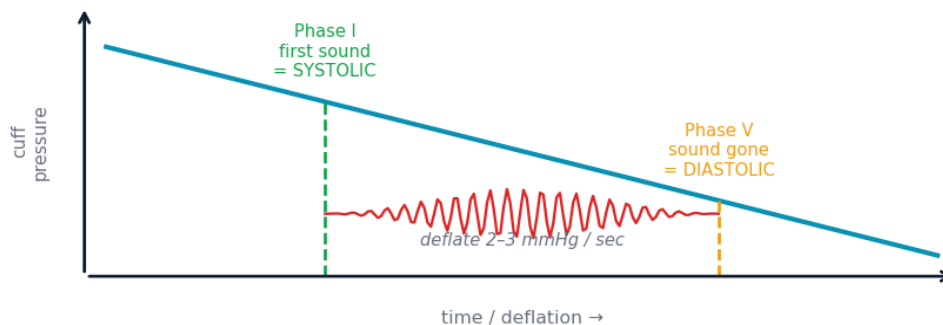
Cuff & Stethoscope Placement



Step 3: Position the Cuff and Stethoscope

Place the cuff on bare skin (not over clothing). Cuff lower edge should be 2–3 cm above the antecubital fossa (elbow crease). The artery marker on the cuff should align over the brachial artery. Place the stethoscope bell (not diaphragm) directly over the brachial artery without going under the cuff. Support the arm at heart level.

Korotkoff Sounds — Inflate & Deflate

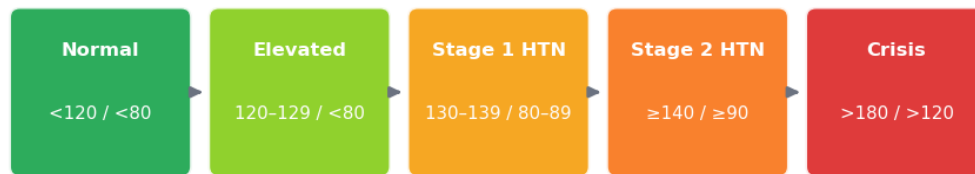


Step 4: Inflate and Deflate

Inflate cuff rapidly to 20–30 mmHg above expected systolic pressure (or 160 mmHg if unknown). Deflate at 2–3 mmHg per second. Listen for Korotkoff sounds: Phase I (first sound heard) = systolic pressure. Phase V (sounds disappear) = diastolic pressure. Record to nearest 2 mmHg.

Note: If you miss the reading, fully deflate for 1–2 minutes before re-inflating.

Blood Pressure Categories (ACC/AHA)



Crisis + symptoms (chest pain, dyspnea, neuro changes) = emergency — call physician.

Step 5: Record and Interpret

Record both readings immediately. For new patients, measure both arms — use the higher reading arm for future measurements. Take two readings 1–2 minutes apart and average them. Normal: <120/80 mmHg. Elevated: 120–129/<80. Stage 1 HTN: 130–139/80–89. Stage 2 HTN: ≥140/≥90. Hypertensive crisis: >180/120.

■ BP >180/120 mmHg with symptoms (chest pain, shortness of breath, neurological changes) = hypertensive emergency — call physician immediately.